

Kisi-Kisi LKS Provinsi 2025:

IT Software Solutions for Business

A. Hardware Recommendation

Tools	Specification
Desktop/Laptop	• Processor 2.0 GHz Quad core or higher • RAM 16 GB or higher • SSD 128 GB or higher • Operating System Windows 10
Mouse	Wired Mouse Optic (No memory or Macro capability allowed)
Keyboard	Wired Keyboard (No memory or Macro capability allowed)
Monitor	20 - 24 inch
Dongle Wi-Fi	For PC User

B. Required Software

Software	Specification
Windows 10, 64 bits	Windows 10 Education 64 Bit
Microsoft Office 365 Home/Education	<i>Free Trial 1 month</i>
Visual Studio Community Edition 2022 (v17.13) • ASP .NET & Web Development ✓ .NET Framework 4.7.X or higher ✓ .NET 8.0 or .NET Core 8.0 • .NET Desktop Development ✓ .NET Framework 4.7.X or higher	-
Microsoft SQL Server Express Edition 2019	-
SQL Server Management Studio 19.x	-
Android Studio Ladybug • Android SDK Tools • Android SDK Platform-Tools • Android SDK Build-Tools • Android SDK Platform • Kotlin	-

• AVD Manager (PIXEL XL API 33) • Android SDK API 33	
Google Chrome Browser (latest version)	-

C. General Requirements:

- **Test project will be given in English. Dictionary (hardcopy) is allowed during the competition.**
- Every module should be provided with proper validation and informative error messages.
- Scrollbar should be visible whenever there are lists or views that occupy more than one screen.
- Use proper naming conventions for variables, functions, or filename.
- By default, form should be displayed in the centre of the screen.
- Whenever a form or dialog is in focus, other form functionality should be disabled.
- Wireframe given in the test project is for reference only. Modifications are allowed if they do not affect the performance and functionality of the requested features.
- Pay attention to given working durations. The time given includes the duration for the submission process. Additional time will not be added during normal conditions.

D. Test Project Detail

Module Name	Duration (Hours)	Requirements
Desktop Application (40%)	3	• Import database and its content from external sources. • Create windows forms and its components based on given requirements. • Validating user input. • Role validation and menu access restriction based on login data.

		• Show, insert, update, or delete records from the database. • Create windows forms and its components based on given requirements. • Join tables query. • Aggregate query. • Display data using charts. • Searching and filtering. • Datagridview manipulation.
Mobile Application (40%)	3	• Create a mobile application form and its component based on given requirements. • Show, insert, update, or delete records from external sources (API). • Searching and filtering. • Export data to external files. • Show Image from others resource. • Animation or micro interactions. • User Authentication and Token Management
Web API (20%)	3	• Analyze given requirements and translate them into an effective database design. • Develop RESTful API endpoints according to the provided specifications. • Implement the appropriate logic for each endpoint. • Ensure each RESTful API endpoint includes correct headers, parameters, request body, method, and suitable return objects (HTTP status code, response headers, and response body). • Handle error scenarios by providing meaningful and consumable responses for all potential negative cases. • Implement search and filtering functionality. • Manage user authentication and token handling securely.

E. Proposed Schedule

Day	Activities	Participant
	08:00 - 12:00 WIB Technical Meeting: 1. Competition Rules and Procedure 2. Technical Description Discussion 3. Test Project Briefing Familiarization: 1. Workshop Setup and Checking 2. Hardware and Software Testing	• Juri • Teknisi • Peserta • Pembimbing (opsional)
	13:00 - 16:00 WIB Competition Day 1 Session 1: Desktop Application	• Juri • Teknisi • Peserta
	08:00 - 11:00 WIB Competition Day 2 Session 1: Mobile Application	• Juri • Teknisi • Peserta
	13:00 - 16:00 WIB Competition Day 2 Session 2: Web API Application	

F. Additional

- Competitors are allowed to bring snacks and beverages.
- Competitors are allowed to bring jackets or medicine (if needed to).
- Competitors are not allowed to listen to any music using headphone/headset
- Competitors are not allowed to make any noises during the competition (including having conversation with other competitors)
- Competitors are not allowed to add any dependencies to Mobile Application project
- Competitors need to bring their own PC/Laptop for the competition.
- Competitors should obey all the requirements that have been given above.

- Competitors must maintain integrity, fairness, and sportsmanship during the competition.